



End Term (Even) Semester Regular/Back Examination May 2026

Roll no.

Name of the Program and semester: B. Tech IVth Semester.

Name of the Course: Design and Analysis of Algorithms

Course Code: TCS 409

Maximum Marks: 100

Time: 3 hours

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

(2X10=20 Marks) -- CO-1

Q1.

- a. Solve the following recurrence relation
 - (i) $T(n) = 4T(n/2) + n^2 \log n$. - using master theorem
 - (ii) $T(n) = T(n-1) + \log n$ - using substitution method
- b. In algorithm design, the Tower of Hanoi problem is a classic example used to illustrate recursion, where a set of disks must be moved from a source peg to a destination peg specific rules.
 - (i) Derive a recursive algorithm to solve the Tower of Hanoi problem for n disks.
 - (ii) Determine the total number of function calls (or moves) required to solve the problem using the recursive approach. Also, analyze the time and space complexity of the recursive solution.
- c. Write an algorithm for Exponential Search. Analyze their time complexities (best, average, and worst cases), and analyze under what conditions Linear, Binary and Exponential search techniques are most efficient.

(2X10=20 Marks)-- CO-2

Q2.

- a. Write the pseudocode for Insertion Sort and explain its working for the given array:
 $A = \{6, 4, 2, 2, 8, 3, 5, 1\}$ also justify insertion sort holds the online, stable and in-place properties.
- b. Design a Backtracking algorithm for the Subset Sum problem. Given a set $\{4, 7, 9, 11, 13, 17\}$ and a target sum = 26, trace the execution of the algorithm and list all possible subsets that satisfy the condition.
- c. Write the pseudocode for the Quick Sort algorithm and explain its working for the array
 $A = \{8, 2, 4, 3, 9, 6, 10\}$. Also, explain how Randomized Quick Sort modifies the pivot selection process and discusses how it helps in improving performance compared to standard Quick Sort.

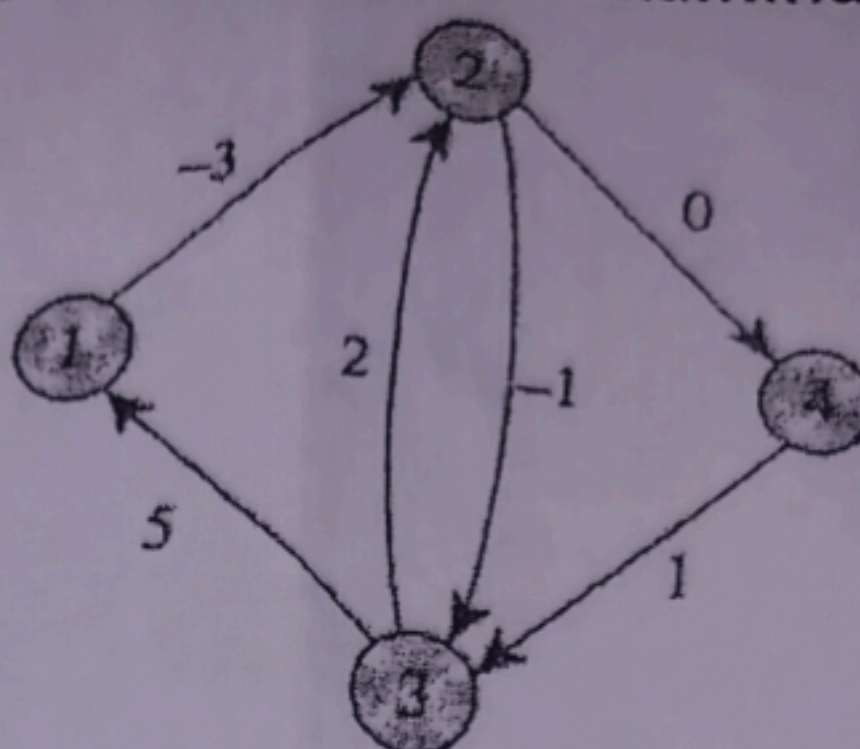
(2X10=20 Marks)-- CO-3

Q3.

- a. Write the pseudocode for Breadth-First Search (BFS) and analyze its time and space complexity.
- b. Write a pseudocode for topological sort and analyze its time and space complexity.
- c. Write the pseudocode for the Floyd-Warshall algorithm to compute the shortest paths between all pairs of vertices in a given below weighted graph and Compute the shortest distance and path matrix step-by-step.



End Term (Even) Semester Regular/Back Examination May 2026



Q4.

(2X10=20 Marks) -- CO-4

- Write an algorithm (pseudocode) to compute the Longest Common Subsequence (LCS) of two given strings and analyze its time complexity. Given the strings: $X = \text{"AGGTAB"}$ and $Y = \text{"GXTXAYB"}$.
- A communication system currently uses 7-bit fixed-length encoding and is being upgraded to Huffman Coding based on symbol probabilities:
 $M1: 0.45$ $M2: 0.02$ $M3: 0.24$ $M4: 0.18$ $M5: 0.11$
 - Construct the Huffman Tree and determine the codes.
 - Decode the message using the generated codes.
 - Calculate the average number of bits per symbol using Huffman coding.
- Given a set of jobs where each job has a deadline and a profit as shown below:

Job	J1	J2	J3	J4	J5	J6
Profit	24	18	22	10	12	30
Deadline	5	3	3	2	4	2

- Determine the optimal sequence of jobs to maximize total profit.
- Calculate the maximum profit obtained.
- Analyze time complexity of the algorithm.

Q5.

(2X10=20 Marks) -- CO-5

- Explain any three commonly used hash functions with suitable examples for handling string keys. A hash function is defined as: $H(\text{key}) = (\text{sum of ASCII values of characters in the key}) \bmod 9$ and linear probing is used for collision resolution. Insert the following string keys into a hash table indexed from 0 to 6: "SUN", "MOON", "STAR", "SKY", "CLOUD", "RAIN", "WIND"
 - Determine the final position of the key "CLOUD" in the hash table.
 - Count the total number of collisions that occur during the insertion process and justify your answer.
- Describe the complexity classes P, NP, and NP-Complete, and explain the relationships among these classes, including how they are interconnected.
- Compute the prefix function π for the pattern $P = \text{abacab}$ using KNUTH-MORRIS-PRATT algorithm